CalibAgent: Task-Aware Active Calibration and Shift Recovery for Quadruped Velocity Commands

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Abstract— **Commercial and learned quadruped controllers often expose a planar velocity command while hiding the dynamics that map commands to realized motion. Payload, surface, controller, and estimator changes can therefore preserve balance yet distort navigation. We present CALIBAGENT, an active calibration layer for this opaque interface. A Bayesian command-to-motion model represents cross-axis coupling and asymmetric response, and a task-weighted integrated variance objective selects commands for a declared deployment distribution. Hard authorization, validation stopping, bounded inverse compensation, and residual-triggered recovery remain separate from acquisition. On 183 passive Unitree Go2 trials, a coupled affine map reduces held-out velocity RMSE by 54.5% relative to raw commands. Across controlled synthetic families, task weighting reaches a joint accuracy–uncertainty target in 18.67 trials, saving 39.5% over Latin hypercube sampling and 18.8% over D-optimal design. A candidate-noise oracle sensitivity changes the heteroscedastic result by one trial without changing the method ranking. Pinned Isaac Lab experiments show 9.3–31.7% held-out RMSE reduction, a matched-budget task-error advantage over D-optimal design on every navigation map, positive early-recovery effects under four shifts with no pre-shift alarms, and failure-aware navigation noninferiority across six held-out maps. These results establish the active mechanism in controlled simulation and the need for a coupled response model in passive hardware data.**

I. INTRODUCTION

A quadruped can remain upright while executing the wrong motion. High-level controllers accept a desired planar velocity $\boldsymbol{u} = [v_x, v_y, \omega_z]^\top$, but the steady body velocity may contain gain error, dead zones, saturation, and cross-axis coupling. These errors are particularly consequential when a planner assumes that the exposed command is a calibrated control variable. Taouil *et al.* showed that learned motion models of a black-box quadruped controller can improve footstep planning [1]; related work has calibrated velocity-controlled kinematic models online for wheeled visual–inertial odometry [2]. The remaining question is how to collect the calibration trials themselves when hardware time is limited and the useful part of command space is task dependent.

Robot calibration has long treated measurement configuration as a design variable [3], [4]. Classical observability and D-optimal criteria prioritize parameter identification [5], [6]. Task-oriented calibration instead scores a design by uncertainty at the end-effector or another quantity of interest [7], [8]. We apply that principle to the external velocity interface of a fixed quadruped controller: trials are chosen for their reduction of predictive uncertainty over commands expected during deployment, not for uniform parameter-space coverage.

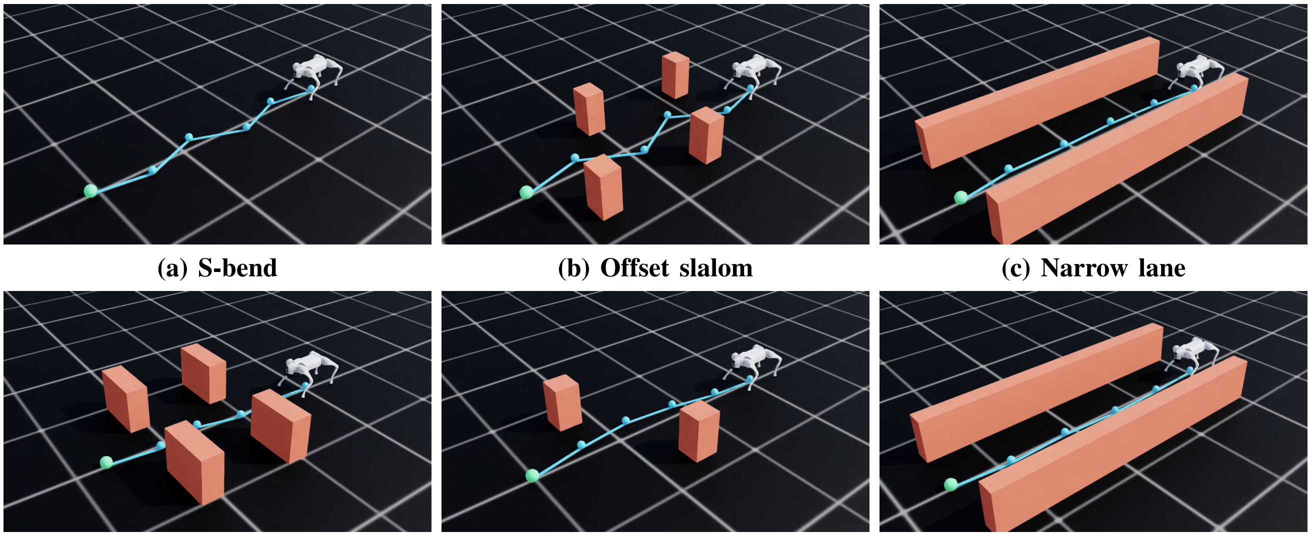
This layer differs from locomotion-policy adaptation. RMA conditions a trained policy on latent environment estimates [9], residual models can update model-based legged controllers online [10], and recent embodiment adaptation identifies hardware changes from short interaction histories [11]. CALIBAGENT neither retrains nor instruments the low-level controller. It identifies the mapping from user commands to measured steady motion, translates desired velocities through a constrained inverse, and reactivates calibration when predictive residuals indicate a shift.

This paper makes four contributions. First, it formulates the opaque command-to-motion interface as a compact Bayesian regression with coupled, asymmetric features and per-trial measurement uncertainty. Second, it gives the conditional one-step task-weighted integrated variance reduction (IVR) score and a safe finite-pool sequential design. Third, it composes acquisition with hard authorization, validation stopping, constrained inversion, and latched shift recovery without allowing statistical value to relax an execution limit. Fourth, it evaluates the full argument from physical model need to downstream navigation using passive Go2 data, controlled synthetic families, fault injection, pinned physics simulation, and held-out navigation across six disjoint maps. The methodological contribution is this task-measure-to-command design and its closed-loop operational composition; the regression, optimal-design update, and change detector remain individually recognizable components. Hardware evidence in this study measures model necessity from passive trials; active selection, recovery, and navigation are evaluated in controlled simulation.

II. RELATED WORK

A. Calibration and Experimental Design

Kinematic and dynamic robot calibration depend on both model structure and the poses or trajectories used for identification [3], [4]. Sun and Hollerbach linked robot observability indices to alphabetic optimal design and developed an efficient active pose-selection algorithm [5], [6]. Carrillo *et al.* then selected configurations by predicted task-space error rather than only parameter covariance [7]. This task-space view parallels goal-oriented Bayesian design, which minimizes uncertainty in a downstream quantity of interest [8]. Greedy information selection also has established connections to near-optimal sensor placement [12]. CALIBAGENT uses a related predictive objective, but its design variable is a safety-filtered velocity command, its integration measure is the planner-facing command distribution, and its posterior is consumed by the same bounded inverse and



(a) S-bend **(b) Offset slalom** **(c) Narrow lane**

(d) Double chicane **(e) Weighted arc** **(f) Extended lane**

Fig. 1. Elevated Isaac Sim views of the six held-out navigation maps. Cyan segments and spheres show the map waypoints; orange geometry shows the obstacles. The overlays are non-colliding visualization geometry and do not contribute to the quantitative endpoints.

change detector. In optimal-design terminology, Eq. (4) is a sequential Bayesian I/V-optimal criterion specialized to a multi-output command interface. Its contribution is the task-measure-to- command specialization, the separation of statistical selection from hard authorization, and the end-to-end calibration–recovery evidence.

B. Legged Command Models and Adaptation

Simulation-trained policies have produced agile and terrain-robust quadruped locomotion [13], [14], [15]. Robust locomotion, however, does not make the user-level velocity interface exact. Learned black-box motion models can bridge that interface for planning [1], while steady-state and local dynamics models can adapt a legged robot to unmodeled disturbances [16]. Policy-level adaptation and domain randomization address broader changes [9], [17], [11]. CalibAgent occupies a narrower operational layer: it actively calibrates a fixed controller's command response, quantifies posterior uncertainty, and applies compensation only inside a non-learned envelope.

III. Method

Figure 2 separates the probabilistic calibration loop from the rules that constrain its operation. The low-level locomotion controller is fixed and treated as a black box. A trial executes a constant body-frame command after warm-up and ramp-in, extracts a robust steady-motion observation, and records its covariance and validity. Invalid observations never update the posterior.

A. Bayesian Command-to-Motion Model

Let $\boldsymbol{y} \in \mathbb{R}^3$ denote the measured steady velocity for command $\boldsymbol{u}$. The coupled affine basis M1 uses $[1, v_x, v_y, \omega_z]$; its diagonal counterpart M0 retains the intercept but zeros every off-axis coefficient. M2 has 13 terms: M1, the three pairwise products, and $[v_k - \tau_k]_+, [-v_k - \tau_k]_+$ for each axis, with $\boldsymbol{\tau} = [0.15, 0.10, 0.25]$. The non-intercept columns are standardized once on a predeclared safe command pool, never on observations or held-out outcomes. The four-term M1 is used for passive hardware model comparison and navigation; M2 is used for synthetic acquisition, closed-loop response, and shift-recovery experiments. Closed-loop fits initialize the linear projection to the identity interface and regularize deviations with a fixed prior. Consequently, the 12-trial navigation fit is a four-parameter-per-output affine update, not an underdetermined M2 fit.

For output axis a, the model is

$$y_a = \boldsymbol{\theta}_a^\top \boldsymbol{\phi}(\boldsymbol{u}) + \epsilon_a, \quad \epsilon_a \sim \mathcal{N}(0, \sigma_a^2 + r_a), \tag{1}$$

where r_a is the measurement-pipeline covariance diagonal and σ_a^2 is the frozen process variance. Given prior $\mathcal{N}(\boldsymbol{m}_{a0}, \boldsymbol{\Sigma}_{a0})$, sequential updates use

$$\boldsymbol{\Lambda}_a \leftarrow \boldsymbol{\Lambda}_a + \frac{\boldsymbol{\phi}\boldsymbol{\phi}^\top}{\sigma_a^2 + r_a}, \tag{2}$$

$$\boldsymbol{\eta}_a \leftarrow \boldsymbol{\eta}_a + \frac{\boldsymbol{\phi} y_a}{\sigma_a^2 + r_a}, \quad \boldsymbol{\Sigma}_a = \boldsymbol{\Lambda}_a^{-1}, \quad \boldsymbol{m}_a = \boldsymbol{\Sigma}_a \boldsymbol{\eta}_a. \tag{3}$$

The predictive mean is $\mu_a(\boldsymbol{u}) = \boldsymbol{m}_a^\top \boldsymbol{\phi}(\boldsymbol{u})$, and the variance for a future noisy observation is $\boldsymbol{\phi}^\top \boldsymbol{\Sigma}_a \boldsymbol{\phi} + \sigma_a^2 + r_a$; reported coverage uses this complete quantity, whereas acquisition uses only the reducible epistemic term. Independent output posteriors keep the online update small while coupled input features represent cross-axis response. Aggregate error uses $\|\boldsymbol{e}\|_W^2 = \boldsymbol{e}^\top W^\top W \boldsymbol{e}$ with $W = \mathrm{diag}((1\,\mathrm{m\,s^{-1}})^{-1}, (1\,\mathrm{m\,s^{-1}})^{-1}, (1\,\mathrm{rad\,s^{-1}})^{-1})$; this is numerically the declared equal-axis convention. Axiswise errors are reported alongside the scalar score.

B. Task-Weighted Active Selection

A declared task distribution is represented by grid commands $\{\boldsymbol{g}_j\}_{j=1}^G$ and normalized weights w_j. For candidate

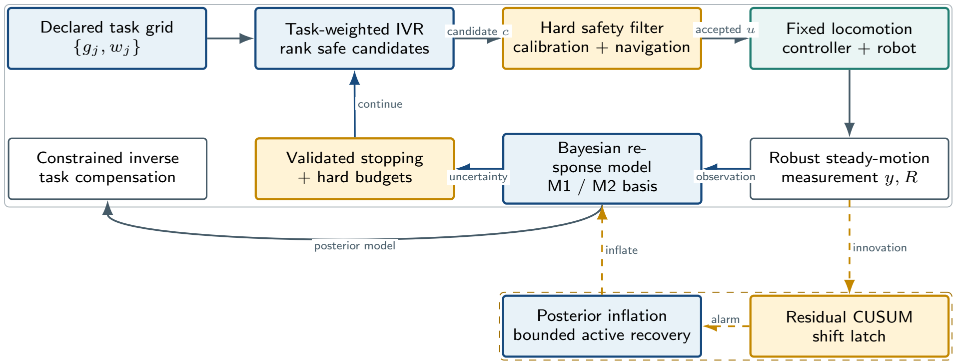


Fig. 2. CALIBAGENT architecture. Task-weighted acquisition ranks calibration commands, but the hard filter alone authorizes execution. Valid observations update the Bayesian model; held-out validation and uncertainty control stopping. A latched residual alarm inflates posterior uncertainty and opens a bounded active-recovery phase. The inverse compensator uses the same bounded command set during navigation.

$\boldsymbol{c}$, let $\tilde{r}_a(\boldsymbol{c})$ denote measurement variance available at planning time. A Sherman–Morrison update gives the outcome-independent, conditional one-step reduction in weighted epistemic variance:

$$\mathcal{I}(\boldsymbol{c}) = \sum_{a=1}^{3}\sum_{j=1}^{G} w_j \frac{\left(\boldsymbol{\phi}(\boldsymbol{g}_j)^\top \boldsymbol{\Sigma}_a \boldsymbol{\phi}(\boldsymbol{c})\right)^2}{\sigma_a^2 + \tilde{r}_a(\boldsymbol{c}) + \boldsymbol{\phi}(\boldsymbol{c})^\top \boldsymbol{\Sigma}_a \boldsymbol{\phi}(\boldsymbol{c})}. \quad (4)$$

The planner sets $\tilde{r}_a = 0$ because a future trial covariance is not observed before execution; Eq. (3) still uses its realized r_a. When a candidate-noise predictor is available, the implementation accepts its nonnegative $\tilde{r}_a(\boldsymbol{c})$. An oracle sensitivity supplies the known synthetic value. The planner maximizes $\mathcal{I}(\boldsymbol{c}) - \lambda_r R(\boldsymbol{c}) - \lambda_d D(\boldsymbol{c})$, where R is normalized command magnitude and D is normalized linear execution distance, over a bounded finite pool; the hard filter authorizes a candidate only after statistical ranking. The sample-efficiency experiment sets both coefficients to zero to isolate IVR. Greedy batches use covariance-only fantasy updates, so no unobserved target is imputed. The synthetic benchmark uses 256 candidates, a six-command seed design, a 256-point task grid, a disjoint 400-point evaluation grid, and a 0.025 normalized duplicate radius. Physics experiments instead use a hard-filtered 128- or 512-command pool and a 0.02 duplicate radius; their task measure is uniform over the held-out command set stated below.

C. Safety, Stopping, Compensation, and Shift Recovery

The hard filter checks finite state, localization, battery, roll, pitch, base height, command bounds, linear norm, coupled linear–angular load, command slew, and projected workspace occupancy. It evaluates ranked candidates after planning and fails closed when none remains. A 50 Hz monitor can zero the command independently of the planner. This separation prevents uncertainty reduction from trading against a learned safety penalty.

Stopping requires minimum trial coverage and measured held-out validation. A run stops at the declared target only after three consecutive checks of both validation RMSE and integrated uncertainty; a low-gain condition also requires valid held-out accuracy. Trial, time, distance, and battery budgets take priority. The synthetic benchmark separately computes an offline oracle crossing from simulator truth; that crossing is an evaluation endpoint and is never visible to acquisition. For deployment, the inverse compensator searches the same bounded candidate set and minimizes

$$\|W(\boldsymbol{\mu}(\boldsymbol{u}) - \boldsymbol{v}^*)\|_2^2 + \lambda\|W(\boldsymbol{u} - \boldsymbol{v}^*)\|_2^2 + \rho\sum_a s_a^2(\boldsymbol{u}), \quad (5)$$

then passes the selected command through the hard filter.

Shift detection uses normalized innovation energy $z_t = \boldsymbol{e}_t^\top \boldsymbol{S}_t^{-1}\boldsymbol{e}_t/3$. A one-sided CUSUM accumulates $q_t = \max(0, q_{t-1} + z_t - 1 - \kappa)$ with $\kappa = 0.5$. It alarms at $q_t \geq 4$ only after three positive-evidence trials within a five-trial window and a three-trial minimum dwell. The alarm latches, inflates epistemic covariance by a factor of eight without changing the posterior mean, and starts a capped recovery budget. Normal stopping resumes after recovery.

IV. Experimental Evaluation

A. Passive Hardware and Sample Efficiency

The hardware dataset contains 183 valid passive Unitree Go2 trials collected in three 61-trial sessions. Livox MID360 FAST-LIO/global localization provides the reference pose. Each session is held out once; a training fold contains 122 trials, of which sampled fits use 30 selected trials and dense fits use all 122. Each command uses 0.6 s ramp-in, 0.8 s settling, 2.0 s measurement, and 0.6 s ramp-out. Held-out three-axis and per-axis RMSE compare raw commands, diagonal affine M0, and coupled affine M1. Two additional nested fits—diagonal linear and coupled linear—separate the effect of an intercept from cross-axis coupling.

Sample efficiency is evaluated over 20 independent seeds in each of three synthetic response families: affine, dead-zone, and heteroscedastic. Every M2 fit begins with the same six-command seed design and has a 160-trial horizon. Candidate, task, and evaluation Sobol sets use independent seeds 31013, 11003, and 23011. The task measure is a Gaussian mixture centered at $(.55, 0, 0)$, $(.30, 0, .70)$, and $(.30, 0, -.70)$ with weights $(.50, .25, .25)$ and diagonal scales

$(.14, .08, .18)$ for the first component and $(.12, .08, .16)$ for the turns. M2 uses prior scale 1.0 and frozen process variances $(.000625, .000324, .001225)$. The 400-point evaluation grid is an offline ground-truth benchmark and never enters acquisition; an independent 1,024-command audit grid (seed 43019) evaluates each posterior at its recorded crossing. The joint target is the first trial with RMSE at most 0.04 and integrated epistemic variance at most 0.0015. Controls comprise random, Latin hypercube (LHS), Sobol, Bayesian D-optimal, no-task IVR, and a 256-command dense oracle. Inference first averages the three response families within each paired seed. The one-sided paired Wilcoxon alternative is fewer trials for task IVR ($n = 20$ seeds; SciPy default zero handling). The LHS contrast is primary; all five baseline comparisons also survive Holm correction ($p_{\mathrm{adj}} \leq 4.77 \times 10^{-6}$). Intervals for trials saved use 5,000 percentile resamples of paired seeds; fixed-budget error intervals use 4,000.

Task-distribution sensitivity freezes every acquisition and re-evaluates the resulting posteriors on a new 1,024-command Sobol grid. Besides the declared mixture, it changes the component weights to forward-heavy $(.8, .1, .1)$, left-heavy $(.1, .8, .1)$, and right-heavy $(.1, .1, .8)$, then replaces the mixture by a uniform measure over the full safe envelope. No command is reselected. This distinguishes robustness to navigation-family reweighting from a support-expanding deployment change.

B. Closed-Loop Simulation and Safety Evaluation

All physics experiments use Isaac Lab v2.3.2 at commit `37ddf62`, Isaac Sim 5.1.0-rc.19, and PhysX [18]. Closed-loop response is measured in four contexts with 20 paired seeds per context. Each M2 fit uses an identity-projection prior scale of 0.01, 12 calibration commands, and eight held-out validation commands. Its warm-up, ramp-in, settling, measurement, and ramp-out phases last 1.0, 0.4, 0.8, 1.0, and 1.0 s, respectively. The common eight-command task/validation measure is uniform over $(-.30, 0, \pm.35)$, $(.20, \pm.18, \pm.30)$ with all four sign combinations, and $(.35, 0, \pm.50)$.

The stopping and authorization tests replay 60 active-acquisition traces, exercise 300 out-of-envelope proposals and 20 safe controls, and inject 160 sampled-state faults. The tested bounds are $|v_x| \leq 0.75$ m/s, $|v_y| \leq 0.45$ m/s, and $|\omega_z| \leq 1.2$ rad/s, with a 20 ms monitoring period. Shift recovery uses 72 paired seeds for each of four held-out in-place changes. Frozen-model, passive-update, and full-recovery arms share 12 initial calibration commands, four no-shift monitor commands, five shifted monitor commands, a 12-command recovery budget, and four-command rolling validation. The early-recovery endpoint, specified before evaluation, averages trials 4–9. A complementary ablation uses 30 paired seeds in each of four held-out shifts and holds the detector, covariance inflation, update rule, safety filter, and 12-command budget fixed while replacing task IVR with no-task IVR, D-optimal, LHS, or random recovery selection. If no ranked candidate is authorized, every adaptive arm executes the same zero-command safe stop. The five trial phases last 0.5, 0.3, 0.6, 0.7, and 0.5 s.

C. Held-Out Navigation

Navigation uses six disjoint maps (Fig. 1), seven calibration methods, and 72 paired seeds per map, totaling 3,024 episodes. All arms share the same controller checkpoint, 50 Hz monitor and predictive interlock, 10 Hz waypoint follower, map geometry, random seed, and planner-configuration hash. The M1 prior scale is 0.10 with process variances $(.0025, .0025, .0050)$; the full method and matched controls use 12 calibration commands, whereas the dense reference uses 30. Eight fixed commands provide calibration validation. Task IVR uses a 512-command safe pool and a uniform measure over $(.12, 0, 0)$, $(.16, 0, 0)$, $(.18, 0, 0)$, $(.14, \pm.08, \pm.15)$, and $(.10, \pm.10, \pm.22)$, where paired signs are matched. The inverse uses regularization .005, posterior-risk weight .01, and axis confidence weights $(1, .5, .5)$.

Safety-system development and navigation inference use disjoint data. On a separate two-map development set, a 10 Hz state guard yielded Full-method success counts of 68/72 and 50/72 and a worst completion-time ratio interval upper bound of 1.347. Trace analysis identified base-height crossings between planner ticks. The resulting 50 Hz predictive height interlock is fixed and shared by every evaluation arm; no development episode enters the reported navigation statistics. The common planner-configuration hash is `9affe07d9231`.

The navigation command envelope is $v_x \in [-0.40, 0.40]$ m/s, $v_y \in [-0.30, 0.30]$ m/s, and $\omega_z \in [-0.70, 0.70]$ rad/s, with linear norm at most 0.45 m/s. Cruise, lateral, and yaw limits are 0.18 m/s, 0.14 m/s, and 0.25 rad/s. Success means entering the 0.25 m goal ball without collision or serious safety event before 60 s. Arrival time is defined only on success; the primary failure-aware time estimand assigns 60 s to every failure. Navigation time excludes calibration, which is reported separately: the deterministic executed-command duration is 31.2 s for a 12-trial method and 78.0 s for dense calibration. Marginal rates use two-sided exact Clopper–Pearson intervals. Paired continuous endpoints, mean ratios, and paired success/collision differences use 4,000 percentile resamples of common seed indices. The navigation margins were fixed at a 1.25 capped-time ratio and a -0.05 success difference: the former limits excess time to 25% of a dense-calibrated traversal, and the latter limits loss to approximately four of 72 episodes when the reference is perfect. An anonymized artifact accompanying the submission contains all immutable configurations, raw metric tables, checksums, evaluators, figure builders, and commands needed to reproduce every reported result.

V. Results

A. Model Need and Calibration Efficiency

Figure 5a shows a consistent held-out benefit from coupled calibration on the passive Go2 data. Pooled RMSE falls from 0.06605 for raw commands to 0.04563 for diagonal-affine

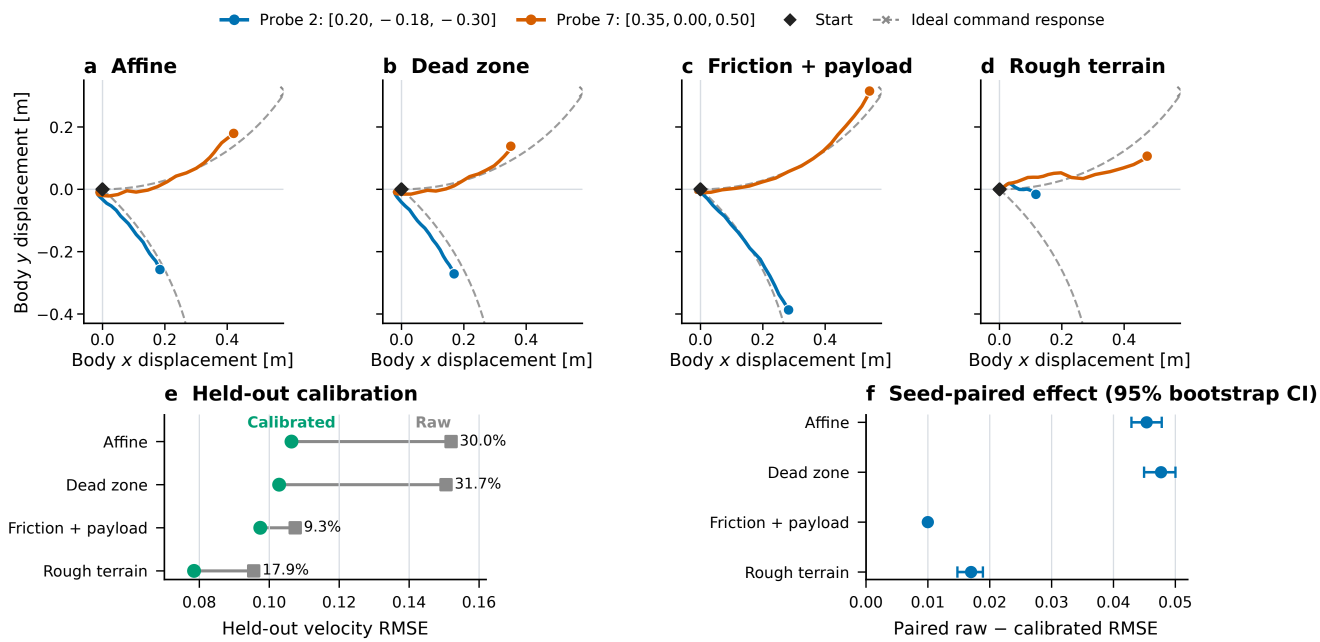


Fig. 3. Closed-loop simulation responses and calibration. (a)–(d) Body-displacement trajectories for two validation probes at display seed 5301; colored curves are measured, dashed gray curves and crosses are the ideal response to the declared command/timing, and diamonds mark the common start. (e) Raw and calibrated held-out velocity RMSE across 20 paired seeds; labels give relative reduction. (f) Seed-paired absolute improvement with 95% 4,000-sample bootstrap intervals. The trajectories are qualitative examples; all inference uses paired summaries specified before evaluation.

M0 and 0.03009 for coupled-affine M1. The nested diagonal-linear and coupled-linear fits obtain 0.04828 and 0.03517, respectively. Coupled linear therefore outperforms diagonal affine even without an intercept, isolating cross-axis response rather than only an offset. M1 therefore reduces error by 54.45% versus raw commands and 34.07% versus M0. The three held-out-session M1 values are 0.02888, 0.03158, and 0.02974. The 30-trial LHS M1 result is 3.33% above the 122-trial dense reference (0.02912). These data support a coupled correction for this robot and acquisition day. Axiswise raw to M1 RMSE is 0.04879 to 0.02587 for v_x, 0.08459 to 0.02569 for v_y, and 0.05960 to 0.03723 for ω_z. The M1-over-M0 gain is concentrated in lateral motion (60.2%); v_x changes by -0.4% and ω_z by 1.0%. The mean learned map (rows are realized axes; columns are command axes) is

$$\bar{A} = \begin{bmatrix} .884 & .015 & -.004 \\ .021 & .653 & .125 \\ -.001 & -.051 & .914 \end{bmatrix}. \tag{6}$$

Across the three complete-session holdouts, the diagonal ranges are [.878,.890], [.629,.667], and [.906,.926]. The yaw-to-lateral term remains .119–.129, while the lateral-to-yaw term ranges from $-.076$ to $-.019$. These data establish passive hardware model need rather than active hardware efficiency.

The synthetic benchmark isolates the acquisition rule (Fig. 5b). The full method reaches the joint target in 18.67 trials, compared with 30.87 for LHS, 30.67 for random, 25.72 for Sobol, 23.00 for D-optimal, and 25.67 without task weighting. Reductions are 39.52%, 39.13%, 27.41%, 18.84%, and 27.27%, respectively; all raw one-sided paired Wilcoxon tests give $p = 9.54 \times 10^{-7}$ and remain significant after Holm correction. The primary LHS comparison saves 12.20 trials with a 95% interval of [10.73, 13.73]. Horizon-end active RMSE is 0.008462, 6.90% above the dense oracle's 0.007916, and future-observation 95% predictive coverage is 94.14%.

The same ordering appears in realized, rather than uncertainty-based, task error (Fig. 5d). At budgets 18, 24, and 30, task IVR obtains mean held-out RMSE 0.02143, 0.01684, and 0.01420. The respective paired improvements are 0.00392, 0.00431, and 0.00466 over D-optimal design and 0.00881, 0.00545, and 0.00598 over no-task IVR; all six 95% intervals exclude zero. At 12 trials, task IVR does not yet separate from those two controls.

Within the navigation task family, the frozen design remains effective under weight mismatch (Fig. 5e). At 24 trials, forward-heavy and right-heavy evaluation preserve positive intervals against all three controls. Left-heavy evaluation preserves advantages over no-task IVR (0.00571, [0.00300, 0.00835]) and LHS (0.00734, [0.00407, 0.01080]); its D-optimal contrast is 0.00202 with an interval crossing zero, but becomes positive by 30 trials. The full-envelope uniform negative control reverses the ordering: task-IVR RMSE is higher by 0.02699 than no-task IVR, 0.02923 than D-optimal, and 0.01421 than LHS at 24 trials. Its error is 3.21 times the declared-task error (Fig. 5f). Task weighting therefore tolerates large mixture reweighting, but a deployment support expansion requires revising the task measure or switching to a general-purpose design.

The uncertainty gate is binding in all 60 active condi-

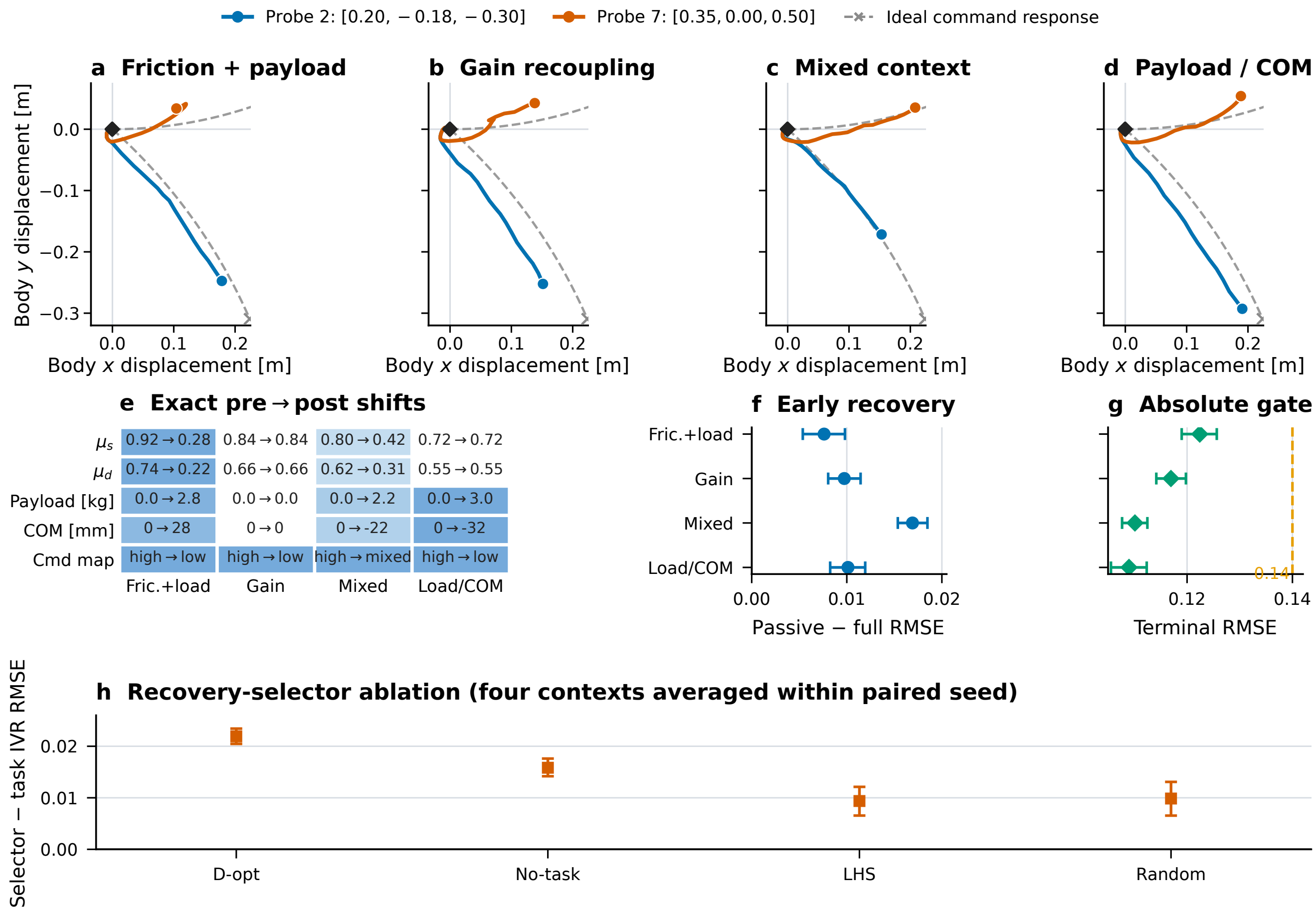


Fig. 4. Held-out simulator shifts and active recovery. (a)–(d) Measured body-displacement trajectories and dashed ideal command responses for the same validation probes at display seed 10101. (e) Exact pre→post static and dynamic friction, payload, fore–aft COM, and command-map settings; color intensity encodes within-row shift magnitude. (f) Full-method improvement over passive updating in the pre-specified early window. (g) Full-method terminal RMSE and the declared 0.14 gate. (h) Recovery-selector ablation, averaging the four contexts within each of 30 paired seeds; positive values favor task IVR. Error bars are paired 95% 4,000-sample bootstrap intervals. The primary recovery experiment uses 72 seeds per shift and produces 0/72 pre-shift alarms in each context (exact upper bound 0.050).

tions, as expected for an uncertainty-directed design; mean accuracy-only crossing occurs at 13.55, 6.40, and 8.80 trials for affine, dead-zone, and heteroscedastic families. Figure 5c therefore reports every family separately. Task IVR requires 18, 18, and 20 trials, versus 19, 19, and 31 for D-optimal and 23, 23, and 31 without task weights. On the disjoint post-hoc audit grid, active posterior RMSE at the primary crossing is 0.01796, 0.02069, and 0.02315; its axiswise 95% coverage ranges from 0.917 to 0.955. Across the nine combinations of RMSE thresholds $\{.03, .04, .05\}$ and uncertainty thresholds $\{.001, .0015, .002\}$, task IVR saves at least 3.33 trials over D-optimal, 6.00 over no-task IVR, and 10.03 over LHS, with a positive paired difference for every seed. Thus the result is not an artifact of a single threshold.

The candidate-noise sensitivity addresses the conditional variance in Eq. (4). In the heteroscedastic family, the primary fixed-noise planner reaches the target in 20 trials; an oracle supplied the exact candidate-dependent excess variance and reaches it in 21. Their crossing RMSEs are 0.02332 and 0.02320. Both retain an 11-trial advantage over D-optimal and no-task IVR. Candidate-noise modeling changes one command in the crossing budget but does not explain the method ranking.

In closed-loop simulation (Fig. 3e), raw-to-calibrated RMSE is 0.15199 to 0.10635 for affine distortion, 0.15054 to 0.10282 for dead-zone distortion, 0.10741 to 0.09742 for low friction plus payload, and 0.09554 to 0.07849 on rough terrain. Relative reductions span 9.30–31.70%. The corresponding paired absolute-improvement intervals are [0.04290, 0.04780], [0.04495, 0.05002], [0.00981, 0.01017], and [0.01477, 0.01889]. All 20 paired seeds improve in each scenario.

B. Stopping, Safety, and Shift Recovery

The stopping test records no premature stop across 60 trajectories. Median and p95 overshoot after the oracle target are both two trials. The hard filter rejects 300/300 declared hazards and 0/20 safe controls; its sampled-state monitor rejects all 160 hazardous calls at the call in which the invalid state is presented. The earlier 0 ms log value is therefore a

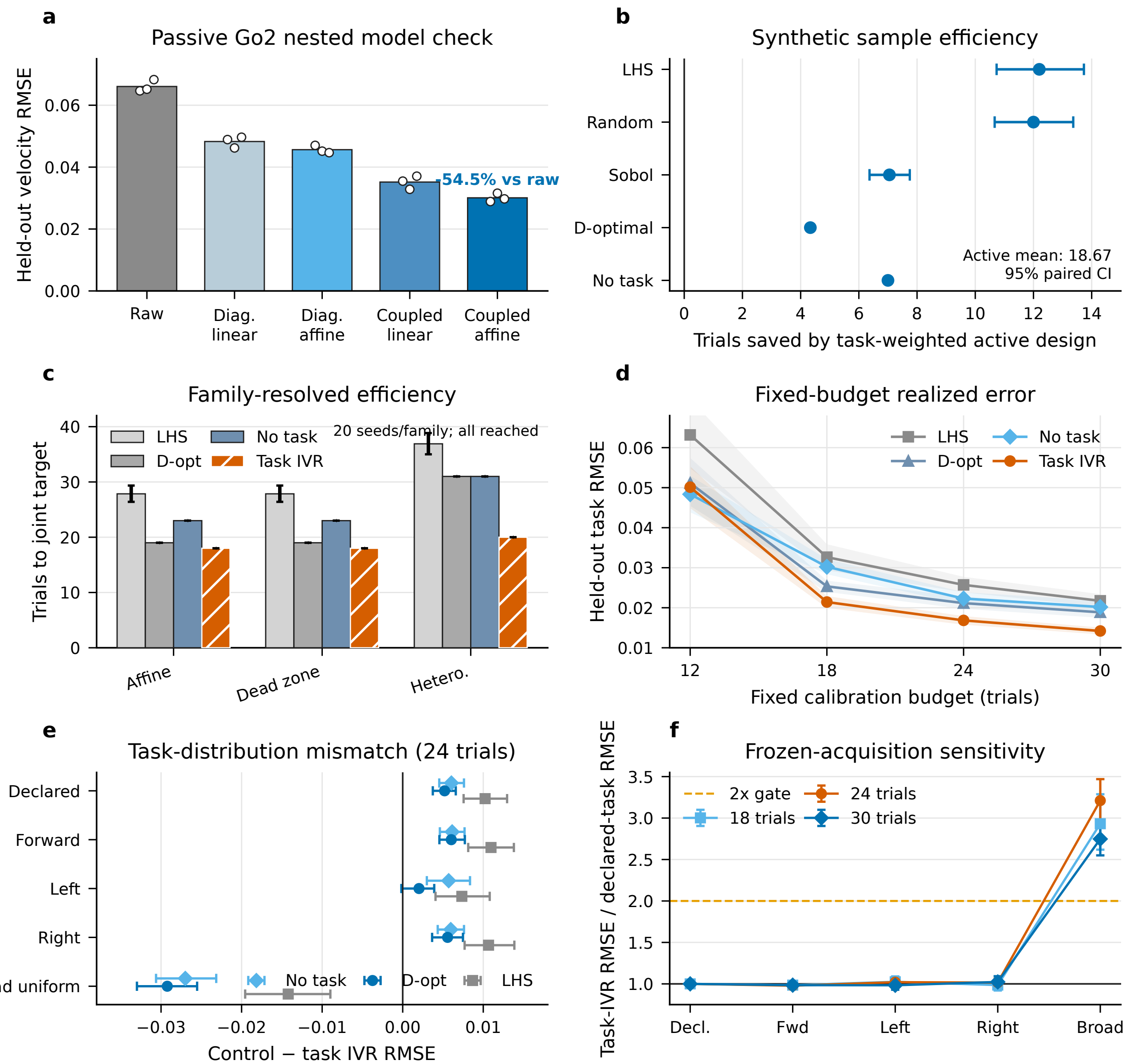


Fig. 5. Calibration evidence. (a) Passive real-Go2 held-out RMSE for raw, diagonal M0, and coupled M1 commands. (b) Paired trials saved by task IVR; bars are 95% seed-bootstrap intervals. (c) Family-resolved time to the same joint target; every method reaches it in all 20 seeds per family. (d) Realized held-out task RMSE at matched trial budgets; bands are 95% paired-seed bootstrap intervals. (e) Frozen-acquisition performance under task-distribution mismatch at 24 trials; positive control-minus-task-IVR values favor task IVR. (f) Task-IVR error relative to the declared measure; the uniform full-envelope negative control crosses the pre-specified 2× sensitivity gate.

software decision timestamp, not a sensor-to-actuator latency measurement. In the physics experiments, where the monitor runs against physics state at 50 Hz, the maximum recorded abort interval is 20 ms. Neither result is physical safety certification.

Across the four no-shift windows, the full detector produces 0/72 alarms per context; each exact 95% upper bound is 0.04994. Across the subsequent shifts, the full method improves early RMSE over passive updating by 0.00761 for friction plus payload, 0.00974 for gain recoupling, 0.01691 for mixed context, and 0.01012 for payload/COM (Fig. 4f). Their 95% intervals are [0.00537, 0.00981], [0.00804, 0.01145], [0.01536, 0.01848], and [0.00825, 0.01194]. Detection succeeds in 72/72 seeds for three shifts and 71/72 for payload/COM; the four-command performance gate is crossed in 72/72 for every shift. This gate is defined by rolling RMSE, not by the alarm: the one payload/COM detector miss crosses it at trial six without a posterior update, so joint detection-and-gate success is 71/72 in that context. Worst p95 detection and recovery delays are four and six trials, corresponding to at most 10.4 s of monitor-command time and 31.2 s for six adaptation-plus-validation cycles under the fixed trial schedule. Full terminal RMSE ranges from 0.10898 to 0.12241, with every upper interval endpoint below 0.126 (Fig. 4g). Passive updating can be competitive at the terminal window, so the supported contrast is the pre-specified early-recovery effect.

The selector ablation isolates the command-selection mechanism (Fig. 4h). After averaging contexts within paired seed, task IVR lowers early RMSE relative to D-optimal by 0.02192 [0.02046, 0.02343], no-task IVR by 0.01583 [0.01418, 0.01761], LHS by 0.00937 [0.00655, 0.01212], and random selection by 0.00986 [0.00654, 0.01309]. It wins all 30 paired seeds against D-optimal and no-task IVR and 27/30 against LHS and random selection. No serious safety event occurs. The early-recovery advantage therefore depends on task-directed command selection, not only detection, covariance inflation, or posterior updating.

C. Navigation Consequence

In the held-out evaluation, Full succeeds in 72/72 episodes on five maps and 70/72 on weighted arc (Fig. 6a), while raw commands succeed in 0/72 or 1/72. Every Full collision count is zero. The primary time estimand retains all episodes by assigning the 60 s cap to every failure; arrival time among successes is only a diagnostic. Paired raw-minus-Full capped-time gains have lower interval endpoints from 23.453 to 32.675 s (Fig. 6b). At the same 12-trial budget, task IVR reduces held-out task-command RMSE over D-optimal design by 9.98–12.28% across all six maps, with every 95% interval above zero (Fig. 6c). The reduction over no-task IVR is 10.32–13.57% on five maps; S-bend gives 2.93% with an interval spanning zero. Thus the physics validation data identify task weighting even where navigation success is near its ceiling. The worst Full/dense mean-ratio interval upper bound is 1.0736, below the declared 1.25 margin, and the largest upper bound against any 12-trial matched-budget control is 1.0901 (reported in the artifact). The paired success-difference lower bound also exceeds the declared -0.05 margin in every comparison. The two Full failures are weighted-arc timeouts, not collisions or serious safety events; their final goal distances are 0.565 and 0.567 m.

Navigation time excludes calibration. Full requires 31.2 s of deterministic command execution versus 78.0 s for dense calibration; matched-budget controls have the same 31.2 s budget. Full mean capped times span 26.17–34.87 s across maps. Its navigation controller invokes 515 regular and nine emergency stall-recovery actions over 432 episodes, with 10 non-serious safety events and no serious event. Identical seeds, policy, planner, interlock, maps, and navigation logic isolate calibration inside this simulator pipeline. The active-hardware protocol tests the corresponding physical navigation claim.

VI. Discussion and Limitations

The experiments separate three questions that are often conflated: whether the exposed command interface needs coupled calibration, whether task weighting changes which trials are valuable, and whether held-out command accuracy preserves navigation outcomes. No-task and D-optimal controls isolate task weighting in the acquisition benchmark; candidate-noise and threshold sweeps preserve that ordering. The mismatch analysis further shows that this benefit survives navigation-family reweighting but is not uniform over an undeclared full-envelope task. This is the intended behavior of a goal-oriented design: the planner must receive an updated task measure when deployment support changes. In navigation, dense and matched-budget controls establish downstream noninferiority, while the raw arm exposes the consequence of leaving the command interface uncalibrated.

The joint benchmark crossing is uncertainty-limited in all 60 active conditions. This identifies the measured efficiency as faster acquisition of posterior confidence, rather than earlier satisfaction of the accuracy gate alone. The independently seeded 400-point evaluation set supplies only the retrospective oracle crossing, and the additional 1,024-point audit verifies RMSE and future-observation coverage at that crossing. A separate trace replay verifies the three-check stopping state machine and its two-trial overshoot; because that replay also uses simulator truth, it tests stopping logic rather than a deployable measurement source. Hardware execution instead obtains the accuracy gate from repeated held-out commands. The shift experiment similarly separates monitoring from recovery: four pre-shift monitor commands measure false alarms, shifted monitor commands measure detection, and trials 4–9 measure the pre-specified early-recovery effect. The null exposure supports the reported per-context false-alarm bound; long-duration nuisance alarms remain a hardware endpoint.

The hardware measurements are passive, same-day data from one Go2 in one environment. Active selection, recovery, and navigation use a pinned simulator and fixed policy. The compact steady-response basis targets the steady command interface; arbitrary history dependence, gait switching, rapid transients, and shifts outside the evaluated families require a richer model. Active hardware evaluation must apply the same calibration and recovery mechanism under gain/coupling, payload/COM, friction, and mixed shifts, with navigation on weighted-arc and offset-slalom courses. This is the remaining evidence needed to establish physical closed-loop performance and sim-to-real transfer.

VII. Conclusion

CalibAgent calibrates the high-level velocity interface of a fixed quadruped controller by directing trials toward task-relevant predictive uncertainty. The approach combines a coupled Bayesian model, conditional IVR acquisition, hard safety filtering, validated stopping, constrained inverse compensation, and latched shift recovery. Passive Go2 measurements establish the need for a coupled map; controlled synthetic and Isaac Lab experiments establish sample efficiency, early shift recovery, and downstream navigation under controlled conditions. Active real-robot execution is required to extend this evidence to physical closed-loop calibration and sim-to-real transfer.

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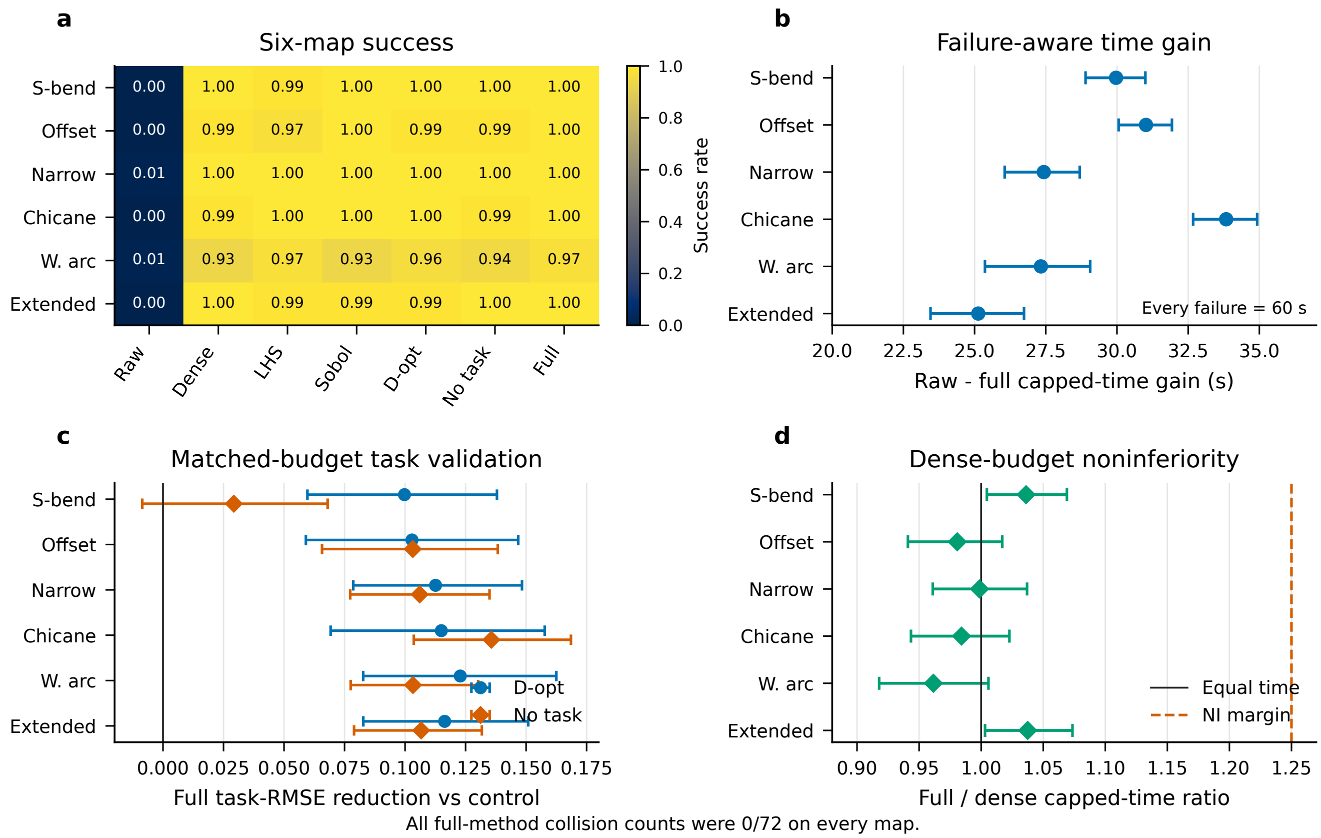

Fig. 6. Held-out simulator navigation (72 seeds/cell). (a) Success rate. (b) Paired raw-minus-full gain in the 60-s capped-time estimand, which retains every failure as 60 s. (c) Matched-budget validation-RMSE reduction from task IVR relative to D-optimal and no-task IVR; bars are paired 95% intervals. (d) Full/dense capped-time ratios and the declared 1.25 noninferiority margin. Every full-method collision count is 0/72.

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